DIGITAL IMAGE PROCESSING LABORATORY

Semester	B.E. Semester VII
Subject	Digital Image Processing
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Laboratory	L05

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Experiment Number	8
Experiment Title	Binary Morphological Processing – Erosion and Dilation
Aim	To convert the segmented eye image of the previous week into a binary image at more than one threshold; to apply **binary erosion** and **binary dilation** with three structuring elements (3x3 square, 3x3 cross, 5x5 square) and study their effect on object size, boundaries, small noise, thin structures, gaps, broken objects and neighbouring objects; to compare the two operations; and to study how repeated application (1, 2 and 3 iterations) shrinks or grows and eventually removes or merges structures. Every effect is **measured** (foreground pixel count, area fraction, connected components, boundary length, the iteration at which regions vanish or merge) rather than described visually only, and the erosion/dilation **duality** is verified numerically. **Lineage.** The input of this experiment is "the final segmented image obtained in the previous week's activity", i.e. the **Task 7 region-growing output**: acquisition → histogram equalization (Task 3) → seeded region growing (`src/region_growing.py`, automatic darkest-point seed, running-mean criterion, T = 45, 8-connectivity – the "High" tolerance of Task 7) → **binary morphology (this report)**. T = 45 is deliberately the loosest Task 7 setting: it also captures the eyebrow and speckle around the iris, which gives the morphological operators real structure (thin bridges, tiny fragments, gaps) to act on; a clean single blob would make erosion/dilation trivial to describe.
Theory	**Binary image and structuring element** A binary image is a set $A \subset \mathbb{Z}^2$ of foreground pixel coordinates (here the dark pupil/iris/eyebrow pixels; background is the complement A^c). A **structuring element** (SE) B is a small set of offsets with a designated **origin**. B_z denotes B translated so its origin sits at pixel z, and the **reflection** is $\hat{B} = \{-b : b \in B\}$. **Erosion (fit)** $$A \ominus B = \{ z : B_z \subset A \} = \cap_{b \in B} A_{-b}$$

A pixel z survives only if the SE centred at z **fits entirely inside** the object. Erosion shrinks objects, strips a layer of boundary pixels, deletes objects and protrusions narrower than the SE, and widens gaps between objects; it can also split an object joined by a thin neck.

Dilation (hit)

$$A \oplus B = \{ z : \hat{B}_z \cap A \neq \varnothing \} = \cup_{b \in B} A_{+b}$$

A pixel z is set if the (reflected) SE centred at z **hits** (overlaps) at least one object pixel. Dilation grows objects, thickens boundaries, closes narrow gaps and holes, joins broken pieces and merges nearby objects. Figure 2 shows both rules on a small shape.

Duality

$$(A \ominus B)^c = A^c \oplus \hat{B} \quad \Leftrightarrow \quad A \ominus B = (A^c \oplus \hat{B})^c$$

Eroding the foreground is the same as dilating the background with the reflected SE, and taking the complement. All three SEs here are symmetric so $\hat{B} = B$, but the code reflects anyway. The identity is verified pixel-for-pixel in Output (Table 10, Figure 17).

Structuring-element geometry

SE size and shape set how much and in which directions an object changes (Figure 1). The **3x3 square** (9 pixels) is 8-connected: it changes an object by one pixel in all eight directions, so diagonal edges move faster (by √2) than axis-aligned ones. The **3x3 cross** (5 pixels: centre + 4 neighbours) is 4-connected: it moves edges by one pixel orthogonally but only ~0.7 pixel along diagonals, so it is gentler and rounds corners into diamonds. The **5x5 square** (25 pixels) is the Minkowski sum of two 3x3 squares, $B_5 = B_3 \oplus B_3$, so one 5x5 pass equals two 3x3 passes (confirmed pixel-identically for both operations on all 3 images). Iterating with the same SE n times equals one operation with the n-fold Minkowski sum: a growing square (or diamond for the cross).

Related composite operations (from `src/morphology.py`)

Opening = erosion then dilation (removes structures smaller than B); closing = dilation then erosion (bridges gaps smaller than B). They are used below as quantitative probes: |A| − |open(A)| counts “thin/small” pixels and |close(A)| − |A| counts bridgeable gap pixels.

Resources / Apparatus Required

Software: Python 3.11 • OpenCV • NumPy • Matplotlib • scikit-image • Pillow • Jupyter Notebook / VS Code. Modules reused: `src/morphology.py`, `src/segment.py`, `src/region_growing.py` (Task 7), `src/enhancement.py` (Task 3). New: `src/binary_morphology.py`.

Algorithm

A. Binary Conversion

1. Read the test image (grayscale; convert with `cvtColor` only if it has 3 channels).
2. Apply histogram equalization (Task 3) and upscale as in Tasks 3–7.
3. Run seeded region growing (Task 7; auto seed, T = 45, 8-connected) to obtain the segmentation; form the **segmented image** S = enhanced pixel inside the region, 255 outside.

4. Threshold: foreground = pixels of S with intensity < t (dark = object), for t ∈ {40, 60, 100}; display original / grayscale / binary and record counts.
5. Select t = 60 as the working binary image for Steps B–D.

B. Erosion

1. Build the SEs: 3x3 square, 3x3 cross (`cv2.MORPH_CROSS`), 5x5 square.
2. For each SE compute A ⊖ B (one iteration).
3. Record foreground count, area fraction, % change, connected components and boundary length; record thin-structure pixels |A| − |open(A)|.

C. Dilation

1. For each SE compute A ⊕ B (one iteration).
2. Record the same measures plus bridgeable-gap pixels |close(A)| − |A|.
3. Compare erosion, dilation and their difference (morphological gradient).

D. Iteration Study and Duality

1. Apply erosion and dilation for 1, 2 and 3 iterations with every SE; tabulate all measures.
2. Keep eroding (up to 150 iterations) until the foreground is empty; record the vanishing iteration of the whole image and of each original component.
3. Keep dilating; record the iteration at which the component count first drops (first merge) and the iteration at which a single component remains.
4. Verify $A \ominus B = (A^c \oplus \hat{B})^c$ and its dual for every image, SE and iteration count, and verify OpenCV against an independent NumPy fit/hit implementation.

Program code

> Trimmed to the essentials. Full module: `src/binary_morphology.py`; driver: `scripts/make_task8_figures.py`. No existing module is modified.

```
def to_binary(img, threshold, dark_foreground=True):
    if img.ndim == 3: img = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
    m = (img < threshold) if dark_foreground else (img >=
threshold)
    return m.astype(np.uint8) * 255

def make_se(shape, size):                       # 'square' or
'cross'
    return morph.structuring_element("rect" if shape == "square"
else shape, size)

def erode(mask, se, iterations=1):  return cv2.erode(mask, se,
iterations=iterations)
def dilate(mask, se, iterations=1): return cv2.dilate(mask, se,
iterations=iterations)

def erode_ref(mask, se, iterations=1):          # A (-) B = AND of
shifted copies
    a = mask > 0; ph, pw = se.shape[0]//2, se.shape[1]//2
    for _ in range(iterations):
        p = np.pad(a, ((ph, ph), (pw, pw)), constant_values=True)
        out = np.ones_like(a)
        for i, j in zip(*np.nonzero(se)):
            out &= p[i:i+a.shape[0], j:j+a.shape[1]]
        a = out
    return a.astype(np.uint8) * 255             # dilate_ref: OR
```

```
of shifts, pad False

def n_components(mask):  return cv2.connectedComponents((mask >
0).astype(np.uint8), connectivity=8)[0] - 1

def boundary_length(mask):                    # fg pixels with a
4-neighbour background
    inner = cv2.erode(mask, cross3,
borderType=cv2.BORDER_CONSTANT, borderValue=0)
    return int(((mask > 0) & (inner == 0)).sum())

def iterations_to_vanish(mask, se, max_iter=150):
    cur = mask
    for k in range(1, max_iter + 1):
        cur = cv2.erode(cur, se)
        if not cur.any(): return k

def merge_iterations(mask, se, max_iter=40):   # first drop in
component count / single component
    n0 = n_components(mask); first = one = None; cur = mask
    for k in range(1, max_iter + 1):
        cur = cv2.dilate(cur, se); n = n_components(cur)
        if first is None and n < n0: first = k
        if n <= 1: one = k; break
    return {"n0": n0, "first_merge_iter": first, "single_iter":
one}

def duality_check(mask, se, iterations=1):
    comp, se_r = 255 - mask, se[::-1, ::-1].copy()
    return int((erode(mask, se, iterations) != 255 - dilate(comp,
se_r, iterations)).sum())
```


Output

Test images and source segmentation (Task 7 lineage)

Label	File	Condition	Upscaled size	Task 7 seed (r,c)	Region px (T=45)	Otsu of S
good_lighting	`alert/s0031_00366_1_0_1_0_1_02.png`	good lighting	256x256	123,114	17523	80.0
poor_lighting	`alert/s0014_08297_0_0_1_1_0_02.png`	poor / uneven lighting	260x260	133,106	11130	88.0
glasses_reflection	`alert/s0035_00133_0_1_1_2_1_02.png`	glasses + IR reflection	288x288	152,143	12683	62.0

> The native images are 128x128, 130x130 and 144x144 and are upscaled 2x (nearest-neighbour) exactly as in Tasks 3–7. The source files are single-channel near-infrared images, so the grayscale-conversion step is a **no-op**: the “grayscale” panel in Figures 3–5 is numerically identical to the original and is shown only to document the step, not because a colour conversion took place.

Diagrams

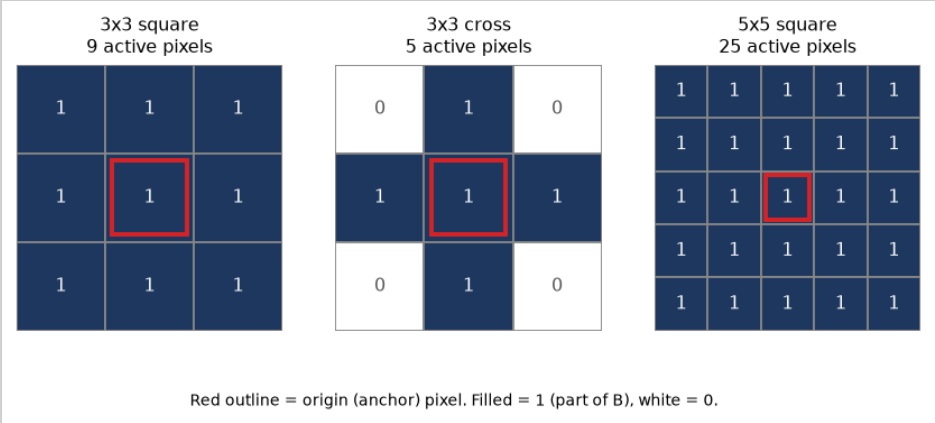


Figure 1: The three structuring elements as pixel grids, origin marked in red.

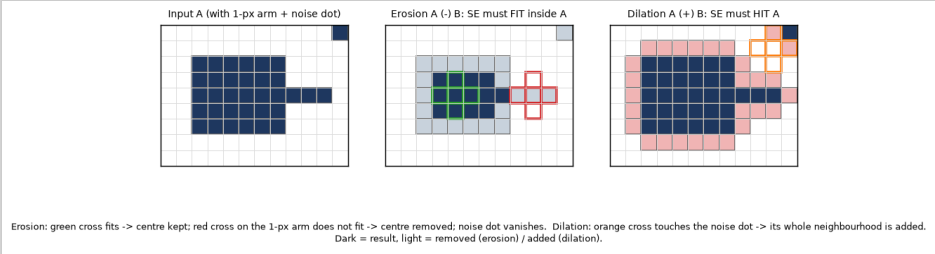


Figure 2: Erosion (fit) and dilation (hit) as a 3x3 cross sliding over a small binary shape.

Task 1 – Binary conversion at three thresholds

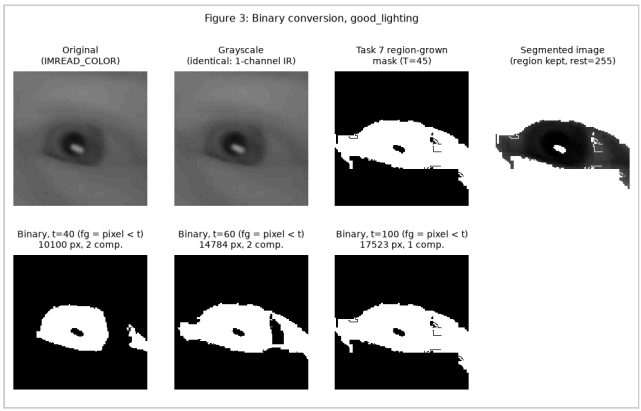


Figure 3: Original, grayscale, region-grown segmentation and binary images at t = 40, 60, 100 (good_lighting).

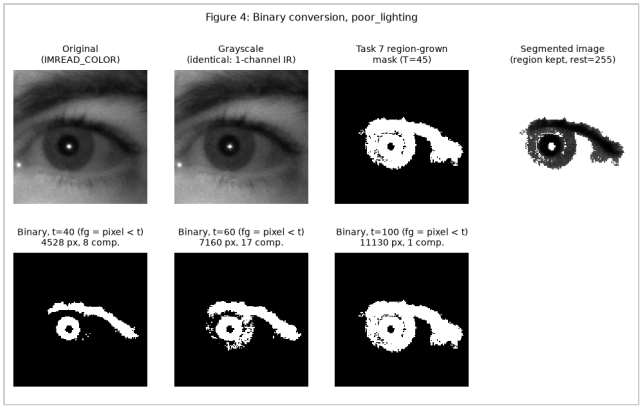


Figure 4: Original, grayscale, region-grown segmentation and binary images at t = 40, 60, 100 (poor_lighting).

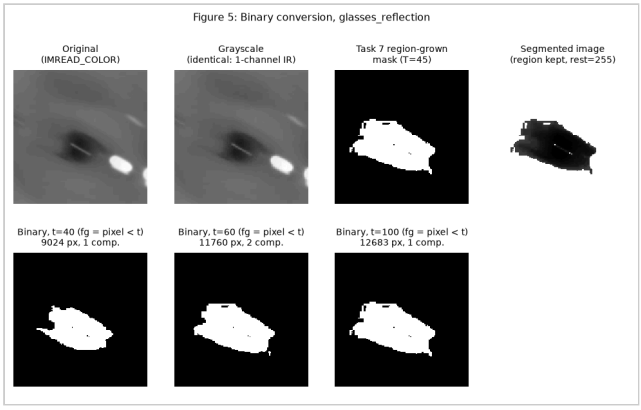


Figure 5: Original, grayscale, region-grown segmentation and binary images at t = 40, 60, 100 (glasses_reflection).

Foreground = pixels of the segmented image S darker than t. Because S already contains only region-grown pixels (values up to about 100 after equalization; background 255), raising t admits progressively brighter boundary pixels of the region:

Image	t	Foreground px	Area fraction	Components	Boundary length	Small comps (≤30 px)
good_lighting	40	10100	0.15411	2	672	0
good_lighting	60	14784	0.22559	2	962	1
good_lighting	100	17523	0.26738	1	1224	0
poor_lighting	40	4528	0.06698	8	773	6
poor_lighting	60	7160	0.10592	17	1417	14
poor_lighting	100	11130	0.16464	1	1467	0
glasses_reflection	40	9024	0.1088	1	533	0
glasses_reflection	60	11760	0.14178	2	612	0
glasses_reflection	100	12683	0.15291	1	695	0

Table 1. Effect of the binarization threshold. Low t keeps only the darkest pupil core (few, separate objects); t = 60 additionally keeps the iris/eyebrow texture as many small fragments (17 components on poor_lighting, 14 of them ≤30 px); t = 100 recovers essentially the whole region-grown mask, which fuses into a single

component on all three images. The Otsu threshold of S (80 / 88 / 62) lies between t = 60 and t = 100. **t = 60 is used for all morphology experiments** because it contains both large objects and genuine small noise/thin structures, which is what erosion and dilation must be judged on.

Working binary images (t = 60):

Image	Foreground px	Area fraction	Components	Boundary length (px)
good_lighting	14784	0.22559	2	962
poor_lighting	7160	0.10592	17	1417
glasses_reflection	11760	0.14178	2	612

Task 2 – Erosion (3 structuring elements, 1–3 iterations)

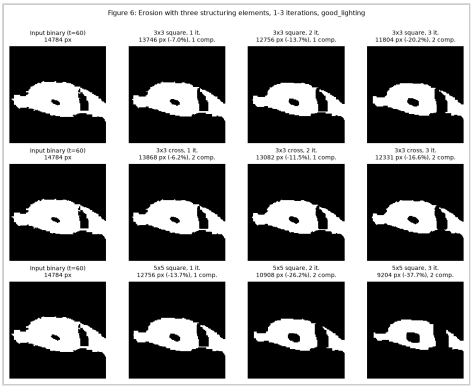


Figure 6: Erosion of the t = 60 binary image, rows = 3x3 square / 3x3 cross / 5x5 square, columns = input and 1–3 iterations (good_lighting).

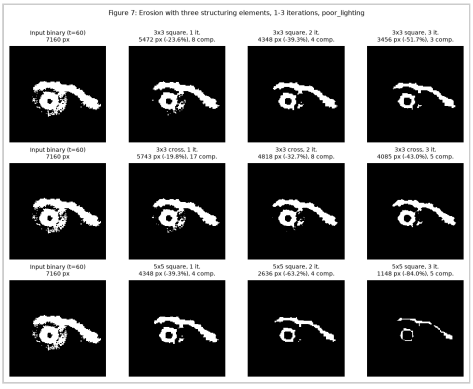


Figure 7: Erosion of the t = 60 binary image, rows = 3x3 square / 3x3 cross / 5x5 square, columns = input and 1–3 iterations (poor_lighting).

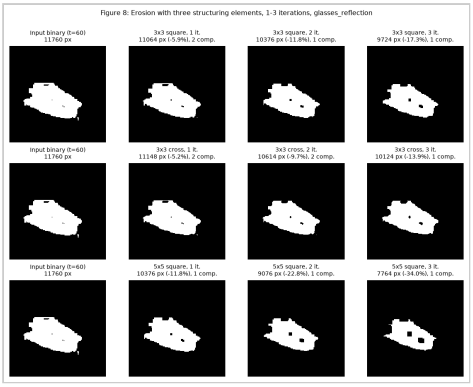


Figure 8: Erosion of the t = 60 binary image, rows = 3x3 square / 3x3 cross / 5x5 square, columns = input and 1–3 iterations (glasses_reflection).

Table 2 – erosion, good_lighting

SE	Iter.	Foreground px	Area fraction	Change vs input	Components (8-conn.)	Boundary length (px)
input	0	14784	0.22559	0.0%	2	962
3x3 square	1	13746	0.20975	-7.0%	1	927
3x3 square	2	12756	0.19464	-13.7%	1	900
3x3 square	3	11804	0.18011	-20.2%	2	844
3x3 cross	1	13868	0.21161	-6.2%	2	830
3x3 cross	2	13082	0.19962	-11.5%	1	793
3x3 cross	3	12331	0.18816	-16.6%	2	734
5x5 square	1	12756	0.19464	-13.7%	1	900
5x5 square	2	10908	0.16644	-26.2%	2	820
5x5 square	3	9204	0.14044	-37.7%	2	761

Table 3 – erosion, poor_lighting

SE	Iter.	Foreground px	Area fraction	Change vs input	Components (8-conn.)	Boundary length (px)
input	0	7160	0.10592	0.0%	17	1417
3x3 square	1	5472	0.08095	-23.6%	8	981
3x3 square	2	4348	0.06432	-39.3%	4	787

3x3 square	3	3456	0.05112	-51.7%	3	728
3x3 cross	1	5743	0.08496	-19.8%	17	925
3x3 cross	2	4818	0.07127	-32.7%	8	733
3x3 cross	3	4085	0.06043	-43.0%	5	646
5x5 square	1	4348	0.06432	-39.3%	4	787
5x5 square	2	2636	0.03899	-63.2%	4	686
5x5 square	3	1148	0.01698	-84.0%	5	564

Table 4 – erosion, glasses_reflection

SE	Iter.	Foreground px	Area fraction	Change vs input	Components (8-conn.)	Boundary length (px)
input	0	11760	0.14178	0.0%	2	612
3x3 square	1	11064	0.13339	-5.9%	2	609
3x3 square	2	10376	0.1251	-11.8%	1	577
3x3 square	3	9724	0.11724	-17.3%	1	576
3x3 cross	1	11148	0.1344	-5.2%	2	534
3x3 cross	2	10614	0.12797	-9.7%	2	490
3x3 cross	3	10124	0.12206	-13.9%	1	477
5x5 square	1	10376	0.1251	-11.8%	1	577
5x5 square	2	9076	0.10942	-22.8%	1	582
5x5 square	3	7764	0.09361	-34.0%	1	596

Task 3 – Dilation (3 structuring elements, 1–3 iterations)

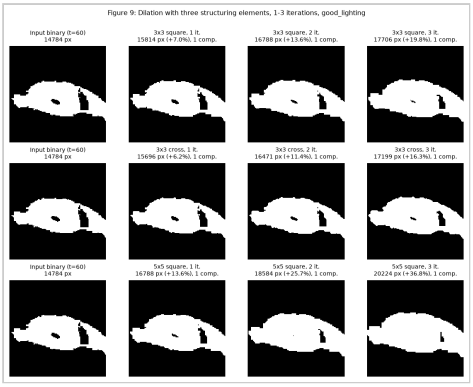


Figure 9: Dilation of the t = 60 binary image, same layout as Figures 6–8 (good_lighting).

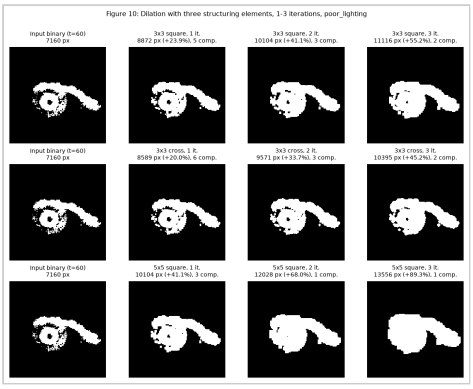


Figure 10: Dilation of the t = 60 binary image, same layout as Figures 6–8 (poor_lighting).

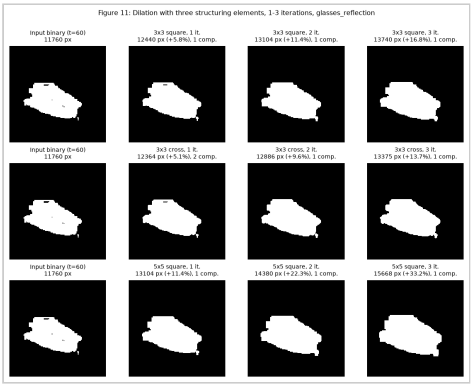


Figure 11: Dilation of the t = 60 binary image, same layout as Figures 6–8 (glasses_reflection).

Table 5 – dilation, good_lighting

SE	Iter.	Foreground px	Area fraction	Change vs input	Components (8-conn.)	Boundary length (px)
input	0	14784	0.22559	0.0%	2	962
3x3 square	1	15814	0.2413	+7.0%	1	930
3x3 square	2	16788	0.25616	+13.6%	1	886
3x3 square	3	17706	0.27017	+19.8%	1	853

3x3 cross	1	15696	0.2395	+6.2%	1	833
3x3 cross	2	16471	0.25133	+11.4%	1	789
3x3 cross	3	17199	0.26244	+16.3%	1	740
5x5 square	1	16788	0.25616	+13.6%	1	886
5x5 square	2	18584	0.28357	+25.7%	1	820
5x5 square	3	20224	0.30859	+36.8%	1	787

Table 6 – dilation, poor_lighting

SE	Iter.	Foreground px	Area fraction	Change vs input	Components (8-conn.)	Boundary length (px)
input	0	7160	0.10592	0.0%	17	1417
3x3 square	1	8872	0.13124	+23.9%	5	1090
3x3 square	2	10104	0.14947	+41.1%	3	899
3x3 square	3	11116	0.16444	+55.2%	2	829
3x3 cross	1	8589	0.12706	+20.0%	6	1010
3x3 cross	2	9571	0.14158	+33.7%	3	863
3x3 cross	3	10395	0.15377	+45.2%	2	733
5x5 square	1	10104	0.14947	+41.1%	3	899
5x5 square	2	12028	0.17793	+68.0%	1	720
5x5 square	3	13556	0.20053	+89.3%	1	645

Table 7 – dilation, glasses_reflection

SE	Iter.	Foreground px	Area fraction	Change vs input	Components (8-conn.)	Boundary length (px)
input	0	11760	0.14178	0.0%	2	612
3x3 square	1	12440	0.14998	+5.8%	1	590
3x3 square	2	13104	0.15799	+11.4%	1	565

3x3 square	3	13740	0.16565	+16.8%	1	570
3x3 cross	1	12364	0.14906	+5.1%	2	518
3x3 cross	2	12886	0.15536	+9.6%	1	485
3x3 cross	3	13375	0.16125	+13.7%	1	473
5x5 square	1	13104	0.15799	+11.4%	1	565
5x5 square	2	14380	0.17337	+22.3%	1	572
5x5 square	3	15668	0.1889	+33.2%	1	581

> Tables 2–7 (erosion then dilation, per image). Boundary length = foreground pixels with a 4-neighbour background pixel. Component counts use 8-connectivity. “Change vs input” is relative to the input binary’s foreground pixel count.

Task 4 – Erosion versus dilation

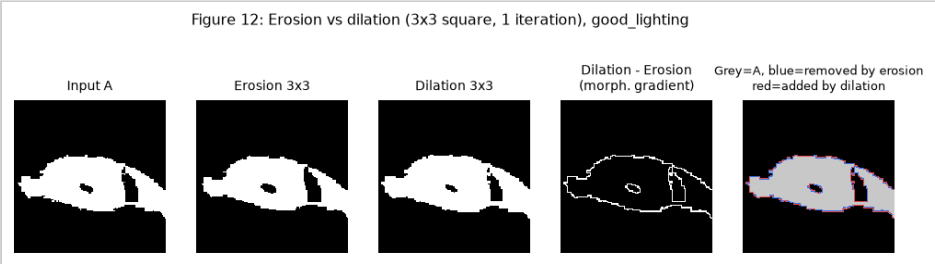


Figure 12: Input, erosion, dilation, morphological gradient and overlay (3x3 square, 1 iteration; good_lighting).

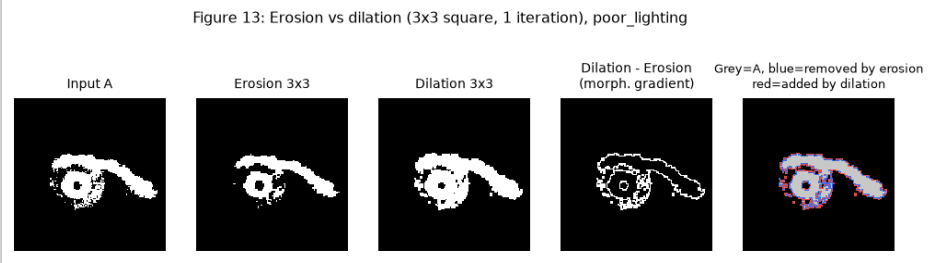


Figure 13: Input, erosion, dilation, morphological gradient and overlay (3x3 square, 1 iteration; poor_lighting).

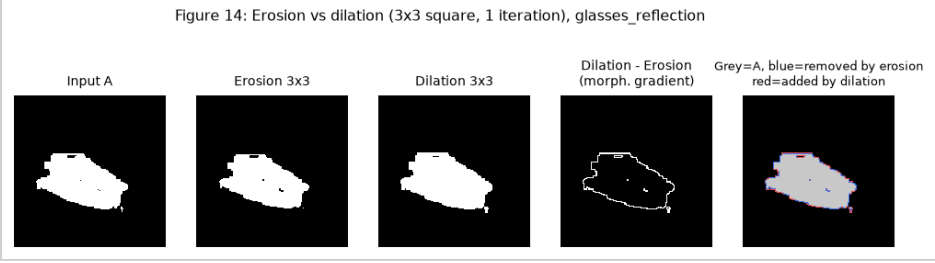


Figure 14: Input, erosion, dilation, morphological gradient and overlay (3x3 square, 1 iteration; glasses_reflection).

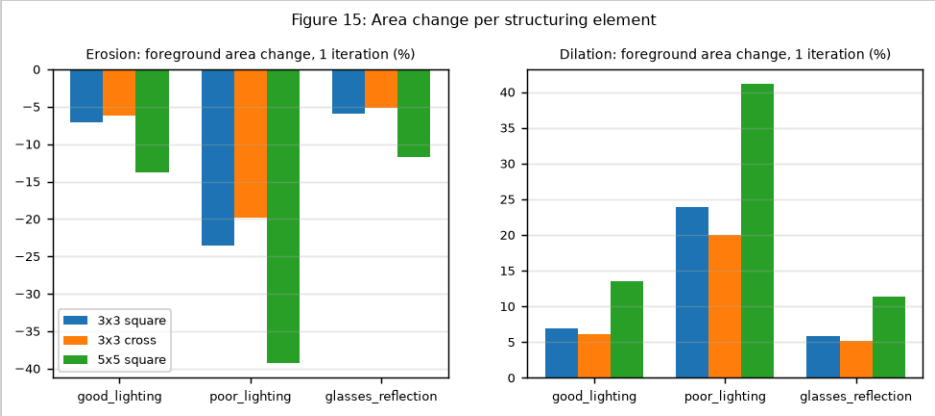


Figure 15: Foreground area change after one iteration, per structuring element and image.

Task 5 – Iterations: vanishing and merging

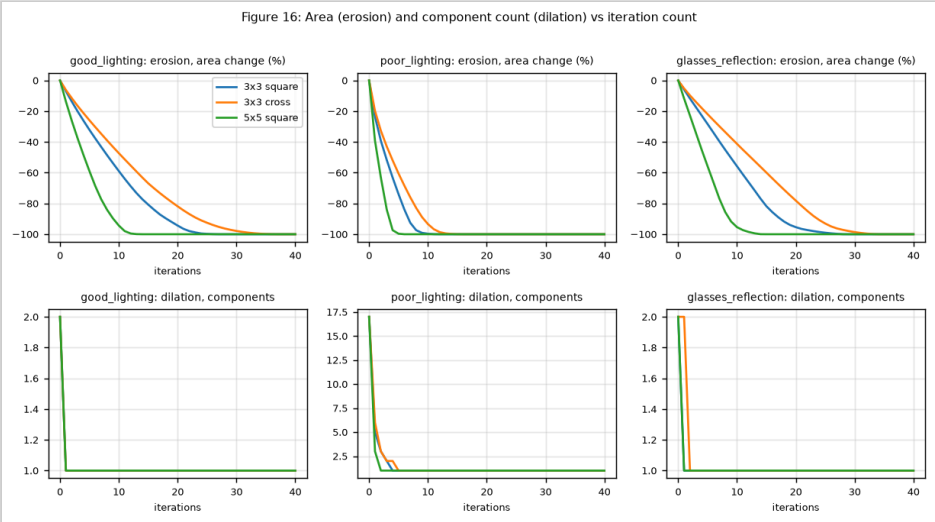


Figure 16: Top: erosion area change (%) versus iterations to 40. Bottom: dilation component count versus iterations.

Image	SE	Initial components	Iteration at which ALL foreground vanishes (erosion)	Vanishing iteration of each original component (erosion)	First merge iteration (dilation: component count drops)	Iteration reaching a single component (dilation)
good_lighting	3x3 square	2	27	1, 27	1	1
good_lighting	3x3 cross	2	37	1, 37	1	1
good_lighting	5x5 square	2	14	1, 14	1	1
poor_lighting	3x3 square	17	12	1 (x15), 2, 12	1	4
poor_lighting	3x3 cross	17	16	1 (x13), 2 (x2), 3, 16	1	5
poor_lighting	5x5 square	17	6	1 (x16), 6	1	2
glasses_reflection	3x3 square	2	29	2, 29	1	1

glasses_reflection	3x3 cross	2	34	3, 34	2	2
glasses_reflection	5x5 square	2	15	1, 15	1	1

Table 8. Vanishing and merging iterations, measured by iterating until the foreground is empty (erosion) or one component remains (dilation).

Thin structures and bridgeable gaps (SE-size probes)

Image	SE	|A| − |open(A)| (thin / small pixels)	|close(A)| − |A| (bridgeable gap pixels)
good_lighting	3x3 square	52	36
good_lighting	3x3 cross	130	127
good_lighting	5x5 square	136	108
poor_lighting	3x3 square	512	448
poor_lighting	3x3 cross	428	419
poor_lighting	5x5 square	952	872
glasses_reflection	3x3 square	16	20
glasses_reflection	3x3 cross	84	86
glasses_reflection	5x5 square	104	96

Table 9. Pixels belonging to structures thinner than the SE (removed by opening) and background gaps narrower than the SE (filled by closing).

Duality verification

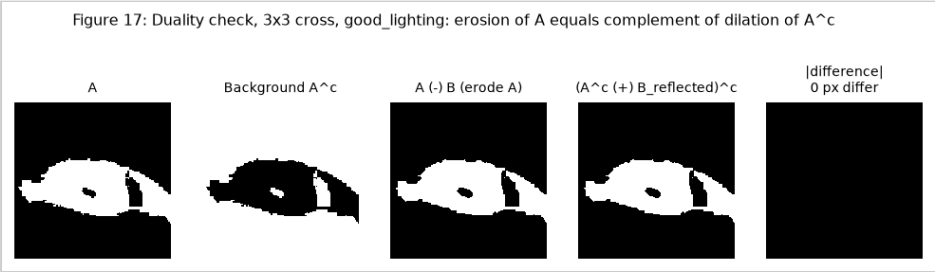


Figure 17: Erosion of A versus complement of dilation of the complement, 3x3 cross, good_lighting; the difference image is empty.

Image	SE	Iterations	$A \ominus B$ vs $(A^c \oplus \hat{B})^c$ differing px	$A \oplus B$ vs $(A^c \ominus \hat{B})^c$ differing px	OpenCV vs NumPy erosion differing px	OpenCV vs NumPy dilation differing px

good_lighting	3x3 square	1, 2, 3	0	0	0	0
good_lighting	3x3 cross	1, 2, 3	0	0	0	0
good_lighting	5x5 square	1, 2, 3	0	0	0	0
poor_lighting	3x3 square	1, 2, 3	0	0	0	0
poor_lighting	3x3 cross	1, 2, 3	0	0	0	0
poor_lighting	5x5 square	1, 2, 3	0	0	0	0
glasses_reflection	3x3 square	1, 2, 3	0	0	0	0
glasses_reflection	3x3 cross	1, 2, 3	0	0	0	0
glasses_reflection	5x5 square	1, 2, 3	0	0	0	0

Table 10. Duality check (differing pixels summed over iterations 1, 2, 3) over 27 cases (3 images × 3 SEs × 3 iteration counts). Total differing pixels: erosion–dilation duality 0, dilation–erosion duality 0, OpenCV-vs-set-definition 0 / 0. **The duality holds exactly (0 differing pixels in every case)**, and the independent NumPy fit/hit implementation reproduces OpenCV exactly. Border handling matters here: OpenCV treats out-of-image pixels as foreground for erosion and background for dilation, which is precisely the dual convention.

Comparison of Erosion and Dilation

Both operators use the same SE and are duals, but they act on opposite sets: erosion trims the object, dilation trims the background. The measured one-iteration changes (Figure 15, Tables 2–7) make the symmetry concrete:

Property	**Erosion A⊖B**	**Dilation A⊕B**
Object size (3x3 square, 1 it.)	good -7.0%, poor -23.6%, glasses -5.9%	good +7.0%, poor +23.9%, glasses +5.8%
Object size (5x5 square, 1 it.)	good -13.7%, poor -39.3%, glasses -11.8%	good +13.6%, poor +41.1%, glasses +11.4%
Boundaries	Move inward; boundary pixel count falls on all images and SEs (e.g. poor 1417 → 981, 3x3 square) as ragged edges and specks disappear	Move outward; boundary pixel count also falls (poor 1417 → 1090) because fragments fuse and notches fill
Small noise	Removed: poor_lighting drops from 17 to 8 components (3x3 square), 4 (5x5); the 32-px dot on glasses_reflection dies at iteration 2	Enlarged and absorbed into neighbours: the same 4-px speck on good_lighting is swallowed by the main object at iteration 1
Thin structures	Eroded away first; the 1-3-px bridges between eyebrow and eye break (good_lighting splits into 2 components at iteration 3)	Thickened; thin strands become solid
Gaps / neighbours	Gaps between objects widen (components can increase)	Gaps close, neighbours merge (components decrease)

Total effect on shape	Anti-extensive: A⊖B ⊂ A (origin in B)	Extensive: A ⊂ A⊕B

The asymmetry between good/glasses (about ±6–7 % per iteration) and poor_lighting (about ±24 %) is a size effect: poor_lighting has a much smaller, more fragmented foreground (7160 px, 1417 boundary pixels) so the boundary layer removed or added is a far larger fraction of it. On the compact good/glasses objects (14 784 and 11 760 px, boundary 962 and 612) the same one-pixel shell is a few percent. A larger SE scales the effect: the 5x5 square changes area about twice as much as the 3x3 square per pass (good -13.7% vs -7.0%), being exactly two 3x3 passes. The cross is the gentlest (good -6.2%, +6.2%) because it does not move diagonal edges.

Erosion followed by dilation (opening) does not return the original: the pixels that stay lost are exactly the structures thinner than the SE (Table 9: up to 952 pixels on poor_lighting with the 5x5 square), and closing bridges gaps by up to 872 pixels. This is why the pair is used together in `src/morphology.py`'s `clean_mask`.

Effect of Iterations

Repeated erosion. Each iteration peels a one-pixel shell (3x3 square), so area falls roughly linearly in iterations for a compact object (Figure 16, top) and the object finally disappears. Measured area change after 1, 2, 3 iterations (3x3 square): good -7.0% / -13.7% / -20.2%; poor -23.6% / -39.3% / -51.7%; glasses -5.9% / -11.8% / -17.3%. With the 5x5 square poor_lighting loses -84.0% after only 3 iterations.

Which structures disappear. Structures vanish in order of width. On poor_lighting (3x3 square) 15 of the 17 original components (the speckle, areas 1–52 px) vanish at iteration 1 and one more at iteration 2; only the main eye/eyebrow object survives, and it sheds small pieces itself (8 components after iteration 1: a 5412-px body plus pieces of 4–20 px). By iteration 3 the body has split into 3 pieces (56, 1084 and 2316 px), the eyebrow band separating from the iris ring (Figure 7). On glasses_reflection the isolated 32-px glare dot vanishes at iteration 2 (3x3 square) or 3 (cross), and on good_lighting the 4-px speck vanishes at iteration 1. The last surviving structure is always the widest part of the pupil/iris region, which vanishes entirely at iteration 27 (good), 12 (poor), 29 (glasses) for the 3x3 square, 37 / 16 / 34 for the cross and 14 / 6 / 15 for the 5x5 square (Table 8). Vanishing time is about half the object's smallest width divided by the SE radius, so the cross (which moves diagonal edges more slowly) lasts longest and the 5x5 square shortest. Erosion can also *increase* the component count by cutting a thin neck: good_lighting goes 2 → 1 → 1 → 2 (3x3 square) because the speck is removed first and the neck to a lower structure breaks at iteration 3.

Repeated dilation. Each iteration adds a one-pixel shell: area grows about +7.0%, +13.6%, +19.8% on good_lighting, but the growth of poor_lighting is far larger (+55.2% after 3 iterations; +89.3% with the 5x5 square) because gaps between fragments are being filled as well as the outline moving outward. The measured component counts fall while the boundary length falls (poor 1417 → 829): the outline becomes smoother.

Which regions merge. The first merge happens at iteration 1 with the 3x3 square, 3x3 cross (poor, good) and 5x5 square on every image; on glasses_reflection the cross needs 2 iterations (the 32-px dot lies diagonally off the main object, and the cross does not extend along diagonals). On poor_lighting

	the 17 components collapse as 5 (iteration 1) → 3 → 2 → 1 for the 3x3 square (single component at iteration 4), 6 → 3 → 2 → 2 → 1 for the cross (iteration 5) and 3 → 1 (iteration 2) for the 5x5 square. The speckle around the iris merges with the iris ring and eyebrow first (after 3 iterations of the 3x3 square only a 64-px fragment remains separate from the 11 052-px main body); the last merge absorbs that residual fragment. Beyond that point further dilation only inflates one merged blob (good_lighting: a single component from iteration 1, area +19.8% at 3 iterations and still growing). **Erosion and dilation are not inverses.** Eroding then dilating with the same SE (opening) restores the large object's shape but not deleted fragments; dilating then eroding (closing) keeps merged objects merged. Repeating either operation is therefore a controlled loss of information, which is the price of noise removal or gap bridging.
Observations	1. All three source images are single-channel infrared, so the grayscale step changes nothing (verified: the panels are numerically identical). 2. The binary image depends strongly on the threshold: for poor_lighting t = 40 / 60 / 100 give 8 / 17 / 1 components and 4528 / 7160 / 11 130 foreground pixels. Thresholds close to the equalized Otsu value (62–88) keep the iris texture as fragments; higher values fill it in. 3. Erosion (3x3 square, 1 iteration) removed 5.9 – 23.6 % of foreground and dilation added 5.8 – 23.9 %; for every image the two are nearly equal and opposite (e.g. good −7.02 % / +6.97 %), differing slightly because of local concavities and image structure. 4. Structuring-element order (least to most aggressive) is 3x3 cross < 3x3 square < 5x5 square for both operators on all 3 images, and the 5x5 square equals two 3x3-square iterations pixel-for-pixel. 5. Erosion removes small noise: poor_lighting component count goes 17 → 8 (3x3 square, iteration 1) and the 32-px glare dot on glasses_reflection is gone by iteration 2. 6. Dilation merges: poor_lighting reaches a single component at iteration 4 (3x3 square), 5 (cross), 2 (5x5). 7. The complete foreground disappears under erosion at iteration 27 / 12 / 29 (3x3 square), 37 / 16 / 34 (cross) and 14 / 6 / 15 (5x5 square) for good / poor / glasses. 8. Boundary length decreases under both operators (smoothing), so it is not a monotone indicator of object size: use area for size and components for noise/merging. 9. The duality $A \ominus B = (A^c \oplus \hat{B})^c$ holds with 0 differing pixels in all 27 cases, for both directions, and OpenCV matches an independent set-definition implementation exactly. 10. Region-grown segmentation at T = 45 includes the eyebrow and speckle around the iris (Task 7 "leakage"); erosion followed by dilation is the natural way to separate the iris from the eyebrow, and dilation/closing the natural way to unify the iris speckle – this is the practical link to the drowsiness pipeline.
Conclusion	Binary erosion and dilation were implemented in `src/binary_morphology.py`, reusing `src/morphology.py` for structuring elements and the Task 7 region-growing output (`src/region_growing.py`) as the segmented input, and

	evaluated on the same 3 dataset images with three thresholds (t = 40, 60, 100; t = 60 used for the experiments), three structuring elements and 1–3 iterations, all with measured pixel counts, component counts and boundary lengths. Erosion shrinks objects, removes noise and thin structures and can split objects; dilation grows objects, closes gaps and merges neighbours. Per iteration the 3x3 square changes area by about 6–24 %, the cross slightly less and the 5x5 square about twice as much; the effect is largest on the small, fragmented poor_lighting foreground. Repeated erosion eliminates the whole foreground after 6–37 iterations depending on image and SE, small fragments first and the iris/pupil body last; repeated dilation merges all fragments into one component within 1–5 iterations. The duality of the two operators was confirmed numerically (0 differing pixels in 27 cases), and an independent fit/hit implementation matched OpenCV exactly. For the drowsiness pipeline the useful operations are the composites: opening to remove speckle and detach the eyebrow, and closing to make the iris a single solid region before the eye-openness measurement.